

Claire Chi

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📍 New York, NY, Open to Relocation Anywhere in the U.S.

Professional Summary

- **7 years of experience** designing, developing, and maintaining scalable enterprise applications using **Java**, **Python**, and modern software engineering practices.
- Strong programming expertise in **Java 8+**, **Python**, and **Object-Oriented Programming (OOP)** concepts including encapsulation, inheritance, polymorphism, and abstraction. Working knowledge of **C++** for algorithm implementation and performance-oriented development.
- Extensive experience building **RESTful APIs**, **microservices**, and distributed systems using **Spring Boot**, **Spring MVC**, **Spring Security**, **Spring Data JPA**, **Hibernate**, and **MyBatis**.
- Strong understanding of **Java 8+** features including **Lambda Expressions**, **Stream API**, **Collections Framework**, **Multithreading**, **CompletableFuture**, and concurrent programming techniques for high-performance applications.
- Experienced in designing **event-driven architectures** using **Apache Kafka**, **RabbitMQ**, and asynchronous processing patterns to support scalable, real-time business workflows.
- Proficient in relational and NoSQL databases including **PostgreSQL**, **MySQL**, **MongoDB**, **Redis**, and **Elasticsearch**, with hands-on experience in query optimization, indexing, caching, performance tuning, and large-scale data processing.
- Experienced in developing and deploying cloud-native applications using **AWS** services including **EC2**, **ECS**, **Fargate**, **EKS**, **S3**, **RDS**, **IAM**, and **CloudWatch**.
- Hands-on experience with **Docker**, **Kubernetes**, **Jenkins**, **Git**, **Maven**, and modern **CI/CD** practices for automated build, testing, deployment, and release management.
- Skilled in implementing secure authentication and authorization solutions using **Spring Security**, **JWT**, **OAuth 2.0**, **SSL/TLS**, and role-based access control mechanisms.
- Frontend development experience using **React**, **Angular**, **Next.js**, **TypeScript**, **JavaScript**, **Redux**, **HTML**, and **CSS** to build responsive and user-friendly web applications.
- Proficient in application monitoring, logging, and observability using **Grafana**, **AWS CloudWatch**, **Log4j**, and distributed system troubleshooting techniques.

- Experienced in **Agile/Scrum** development environments, collaborating with cross-functional teams to deliver high-quality software solutions through iterative development and continuous improvement.
- Hands-on experience integrating **LLM APIs, Retrieval-Augmented Generation (RAG), Vector Search, AI Agent Workflows**, and intelligent automation solutions to improve operational efficiency and information accessibility.
- Strong analytical, debugging, and problem-solving skills with a focus on **scalability, performance optimization, system reliability**, and maintainable software design.

Technical Skills

Programming Languages	Java (Java 8/11, JDBC, JPA, JUnit, Multithreading, Lambda Expressions, Stream API), Python (Pandas, NumPy, Scikit-Learn, OpenAI API), JavaScript (ES6+), TypeScript, SQL, C# (.NET Core, ASP.NET MVC, Entity Framework), C++, PowerShell
Frameworks & Technologies	Spring Boot, Spring Framework, Spring MVC, Spring Security, Spring Data JPA, Spring AOP, Spring Batch, Spring Cloud, Hibernate, MyBatis, Java EE (J2EE), REST APIs, FastAPI, Microservices, React, Angular, Next.js, React Native, Redux
App Services	Apache Tomcat, Jetty, Nginx, JBOSS, IIS
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB, Cassandra, Oracle
Cloud & DevOps	AWS (Lambda, RDS, SNS, S3, SQS, CloudWatch, API Gateway, EC2, Route 53, MSK, Auto Scaling), Azure, Docker, Kubernetes
Big Data & Message Queue	RabbitMQ, Redis, Cassandra, Elasticsearch, Apache Kafka, Apache Spark, WebSocket
Logging & Monitoring	Log4J, Datadog, AWS CloudWatch, Grafana, Splunk, Zipkin
CI/CD Tools	Jenkins, Docker, GitHub Actions, Git, AWS Code Pipeline, Terraform, Azure DevOps, Maven, Gradle, JIRA, TFS
Security	Spring Security, OAuth 2.0, JWT, SSL/TLS, SAML, Okta, Threat Modeling, OWASP Top 10, API Security, Secure SDLC
Operating systems	Windows, Linux, Unix
AI / Machine Learning	OpenAI API, Retrieval-Augmented Generation (RAG), Vector Search, AI

	Agent Workflows, Prompt Engineering, TensorFlow, PyTorch, Keras, Scikit-Learn, Machine Learning, Classification, Regression, Clustering, A/B Testing
Mathematics	Linear Algebra, Discrete Math, Statistics

Work History

Flexport – Java Full Stack Developer

New York, NY

08/2024 - 06/2026

Project Description: Developed and enhanced a cloud-based logistics management platform that enabled real-time shipment tracking, warehouse operations, customer notifications, and document management for global supply chain workflows. The platform provided shipment visibility, logistics event monitoring, and operational reporting through scalable microservices and RESTful APIs. To improve operational efficiency, an AI-powered logistics assistant was introduced to help internal teams quickly access shipment information, summarize logistics events, and retrieve relevant operational documentation through intelligent search capabilities. Leveraging AWS cloud services, Kafka-driven event processing, Elasticsearch, and containerized microservices deployed on Kubernetes, the platform supported high availability, scalability, and efficient handling of large volumes of logistics transactions and shipment events.

Responsibilities:

- **Agile Development:** Participated in Agile ceremonies including sprint planning, daily stand-ups, backlog grooming, and retrospectives to support continuous delivery and cross-functional collaboration.
- **REST API Development:** Designed and developed scalable **RESTful APIs** using **Java 11**, **Spring Boot**, and **Spring Data JPA** to support shipment tracking, order management, warehouse operations, and logistics reporting.
- **Microservices Architecture:** Built and maintained distributed **microservices** responsible for shipment lifecycle management, event processing, document retrieval, and notification delivery.
- **Business Logic Implementation:** Implemented core logistics workflows for shipment status updates, route management, exception handling, and operational reporting using **Spring Boot** and **Hibernate**.
- **Event-Driven Processing:** Developed asynchronous services using **Apache Kafka** to process shipment events and deliver real-time updates across multiple downstream systems.
- **Data Management & Search:** Designed and optimized data persistence solutions using **PostgreSQL**, **MongoDB**, **Redis**, **Elasticsearch**, and **Spring Data JPA** to support transactional workloads, high-performance search, and low-latency data access.
- **Cloud & Containerization:** Leveraged **AWS** services including **EC2**, **ECS Fargate**, **EKS**, **S3**, **RDS**, **IAM**, and **CloudWatch** while utilizing **Docker** and **Kubernetes** to deploy scalable, highly available cloud-native applications.
- **Frontend Development:** Developed responsive user interfaces using **Angular**, **React**, **TypeScript**, **HTML**, and **CSS** for

shipment visibility dashboards, operational tools, and document management workflows.

- **Authentication & Security:** Implemented secure authentication and authorization mechanisms using **Spring Security**, **JWT**, and **OAuth 2.0** to protect APIs and sensitive business data.
- **CI/CD Automation:** Built and maintained automated deployment pipelines using **Jenkins**, **Maven**, **Git**, and **GitHub**, improving release efficiency and deployment reliability.
- **Monitoring & Observability:** Configured **Grafana**, **AWS CloudWatch**, and **Log4j** to support application monitoring, log analysis, alerting, and production troubleshooting.
- **AI Agent Integration:** Developed AI-powered logistics assistants using **LLM APIs**, **Prompt Engineering**, **Retrieval-Augmented Generation (RAG)**, **Vector Search** and **Semantic Search** capabilities to automate operational workflows, summarize shipment events, retrieve logistics documentation, and provide contextual recommendations for operations teams.
- **Agent Orchestration Services:** Designed and developed backend orchestration services using **Spring Boot** and **Python** to integrate AI agents with shipment tracking systems, document repositories, notification services, and operational workflows. Implemented document ingestion, embedding generation, retrieval pipelines, and context enrichment to support intelligent search and question-answering capabilities.
- **Testing & Quality Assurance:** Developed automated unit and integration tests using **JUnit**, **Mockito**, and **Test-Driven Development (TDD)** practices to ensure application stability and code quality.

Tech Stack:

Java 11/17, Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, React, TypeScript, PostgreSQL, MongoDB, Redis, Elasticsearch, Apache Kafka, Docker, Kubernetes, AWS (EC2, ECS Fargate, EKS, S3, RDS, IAM, CloudWatch), Jenkins, Maven, Git, JUnit, Mockito, Grafana, Log4j, JWT, OAuth 2.0, OpenAI API, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Search, AI Agent Workflows, Python, Agile/Scrum, JIRA

Phreesia - Java Backend Developer

New York, NY

08/2023 - 07/2024

Project Description:

Developed and enhanced a cloud-based patient engagement and appointment management platform used by healthcare providers to streamline patient scheduling, digital intake forms, appointment notifications, and insurance verification workflows. The platform processed large volumes of patient interactions and healthcare records through scalable microservices and secure APIs. The solution leveraged event-driven architecture, cloud infrastructure, and search capabilities to improve operational efficiency, reduce administrative workload, and provide a seamless experience for both healthcare staff and patients.

Responsibilities

- **Backend Service Development:** Designed and developed scalable **RESTful APIs** using **Java 11**, **Spring Boot**, and **Spring Data JPA** to support patient scheduling, appointment management, insurance verification, and notification workflows.

- **Microservices Architecture:** Built and maintained **Spring Boot microservices** responsible for patient records, appointment processing, eligibility verification, and communication services.
- **Business Logic Implementation:** Implemented healthcare workflow logic including appointment scheduling, patient check-in, reminder processing, and insurance validation.
- **Data Management & Search:** Designed and optimized **PostgreSQL** schemas, integrated **Redis** caching and **Elasticsearch** search capabilities to improve application performance and information retrieval.
- **Event Processing:** Developed **Kafka**-based event processing services to support appointment scheduling events, notification delivery, and asynchronous workflow integrations.
- **Cloud & Containerization:** Leveraged **AWS (EC2, ECS Fargate, RDS, S3, IAM, CloudWatch)** and **Docker** to deploy scalable backend services in highly available environments.
- **Security Implementation:** Applied **Spring Security, JWT, OAuth 2.0**, and API security best practices to protect sensitive healthcare information and ensure secure system integrations.
- **CI/CD Automation:** Developed automated build and deployment pipelines using **Jenkins, Maven, Git, and GitHub** to improve release efficiency.
- **Monitoring & Logging:** Configured **CloudWatch, Grafana, and Log4j** to support system monitoring, troubleshooting, and production support activities.
- **Data Automation:** Developed **Python scripts** to automate operational reporting, data validation, and batch processing tasks, reducing manual effort and improving reporting accuracy.
- **Testing & Quality Assurance:** Created unit and integration tests using **JUnit** and **Mockito**, improving application reliability and maintaining code quality standards.

Tech Stack

Java 11, Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, PostgreSQL, Redis, Elasticsearch, Apache Kafka, Docker, AWS (EC2, RDS, S3, CloudWatch), Jenkins, Maven, Git, GitHub, JUnit, Mockito, JWT, OAuth 2.0, Python, Agile/Scrum, JIRA

Fxiaohe - Software Engineer

Beijing, China

01/2020 - 06/2023

Project Description

Contributed to the development and enhancement of a cloud-based Customer Relationship Management (CRM) platform used by sales teams and business organizations to manage customer information, sales pipelines, activity tracking, task management, and operational reporting. The platform provided centralized customer data management, automated business workflows, and real-time notifications to improve sales efficiency and collaboration across departments. Participated in backend service development, database optimization, system integrations, and production support activities. The platform supported thousands of business users and processed large volumes of customer interactions, sales records, and operational data through scalable Java-based services.

Responsibilities

- **Backend Development:** Developed and maintained backend services using **Java 8**, **Spring Boot**, and **Spring MVC** to support customer management, sales workflows, activity tracking, and reporting functions.
- **REST API Development:** Designed and implemented **RESTful APIs** for customer records, lead management, sales opportunities, and business process automation.
- **Data Management & Optimization:** Developed database access layers using **MyBatis** and **MySQL**, optimized SQL queries and indexes, and implemented **Redis** caching to improve application performance and reduce database workload.
- **Message Queue Integration:** Implemented asynchronous processing using **RabbitMQ** to support notification delivery, workflow events, background tasks, and business process automation.
- **Business Workflow Automation:** Participated in the implementation of automated business processes including customer assignment, follow-up reminders, task notifications, and approval workflows.
- **Third-Party Integration:** Integrated external services such as email notifications, SMS services, and authentication systems through REST-based interfaces.
- **Data Processing Optimization:** Contributed to performance-sensitive data processing modules using **Java** and **C++** to support reporting workflows and large-scale data operations.
- **System Maintenance & Bug Fixing:** Investigated and resolved production issues, analyzed application logs, and performed root-cause analysis to improve system stability.
- **Testing & Code Quality:** Developed automated tests using **JUnit**, participated in code reviews, and maintained technical documentation to improve application reliability and development standards.
- **CI/CD Support:** Participated in application deployment and release activities using **Jenkins**, **Maven**, and **Git**, supporting continuous integration and delivery practices.
- **Monitoring & Troubleshooting:** Utilized application logs and monitoring tools to identify performance bottlenecks and support production troubleshooting.

Tech Stack

Java 8, Spring Boot, Spring MVC, MyBatis, MySQL, Redis, RabbitMQ, REST APIs, Maven, Jenkins, Git, Linux, JUnit, Apache Tomcat, JSON, C++, Agile/Scrum, JIRA

Education

Cornell University - Ithaca, NY | Master of Science

Computer and Information Systems

Beijing Normal University - Beijing, China | Bachelor of Science

Geographic Information Science